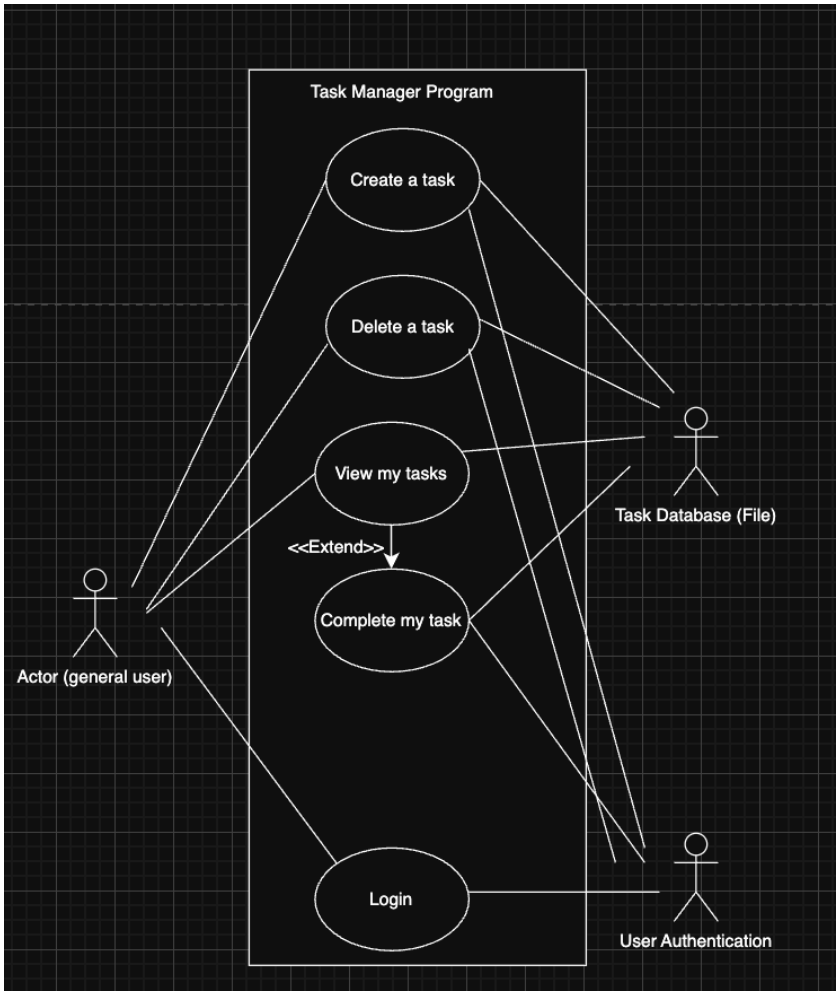
Designing a Task Manager Application Task

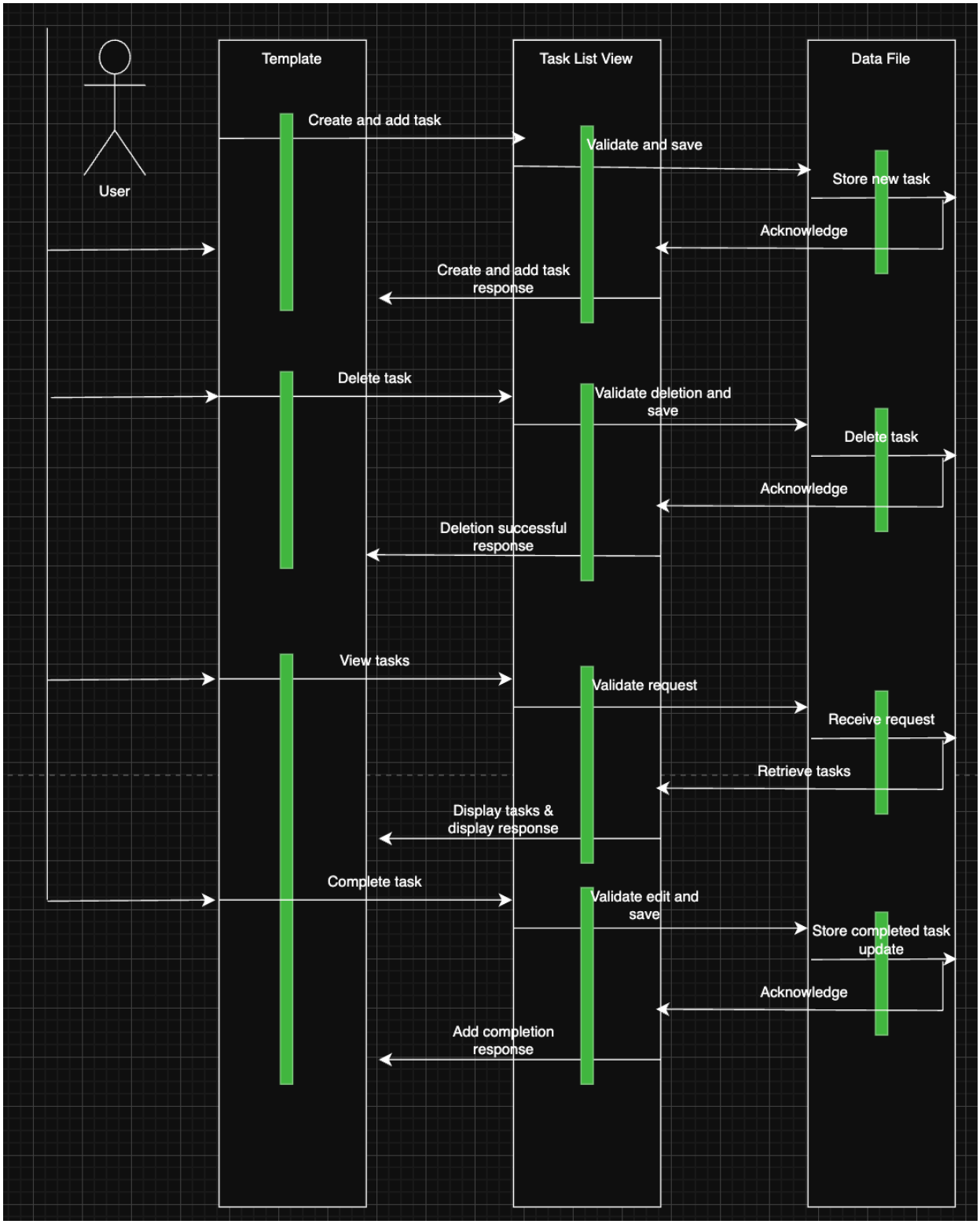
1. Use case diagram

CRUD:

- Create a task.
- Read would be viewing all tasks.
- Update would be editing the task details (ex. Marking as completed which is an extended relationship as it is optional when viewing the tasks).
- Delete would be deleting a task from the list.



2. Sequence diagram



3. MVC

Model:

a. This would define what a task is (ex. title, description, completion status).

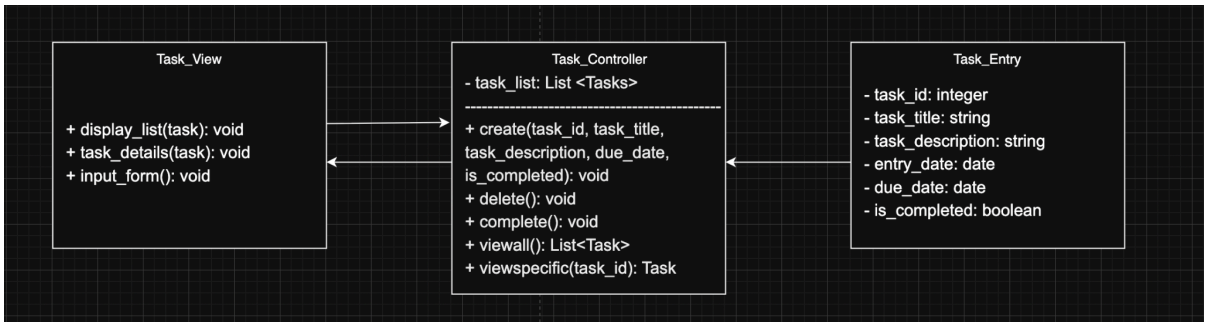
View:

a. This would display the task manager application to the user.
b. It would provide the user with the means to create a task, delete and update a task.
c. It would show the user responses when their action is successful (ex. Task added successfully!).

Controller:

a. Ensures communication between the view and the model (ex. When the user clicks on view my tasks, it would send this request to the model to display all tasks and then it would send the message back to the view to render this new task view to the user).
b. Would update task completion (ex. Incomplete to complete).
c. Contain all the classes and methods to retrieve the file data from (ex. create_task() to save a new task to the file).
d. Validates the user inputs (ex. The user has to enter a task title).

4. Class model



Task Entry Class

This would handle the task objects.

Attributes:

task_id: integer (private)
task_title: string (private)
task_description: string (private)
entry_date: date (private)
due_date: date (private)
is_completed: boolean (private)

Task_View Class

This class would manage what the user sees.

Methods:
display_list(task) - shows all the tasks
task_details(task) - allows a user to see a specific task
input_form() - shows a form to create a new task

Task_Controller Class

This class would control the specific requests made between model and view.

task_list - list of all tasks.
create(task_id, task_title, task_description, due_date, is_completed) - validates user input and calls on save().
delete() - calls on delete() and shows the user's view without the deleted task.
complete() - gets the task and changes the is_completed boolean to true and updates this.
viewall() - receives the request to view all tasks and requests this from the file.
viewspecific() - receives the request to view all tasks and requests this from the file.

5. CRUD Matrix

Entity	Create	Read	Update	Delete
Task Entry	X	X	X	X

- The entity is a **Task Entry** which is the data object.
- A **Task Entry** would have the title, description, entry_date, due_date, is_completed.
- Create: creates a new task entry.
- Read: would be viewing all task entries and also viewing information about a specific task entry.
- Update: would be editing the task entry details (ex. Completing a task).
- Delete: would be deleting a task entry from the file.